Two endpoint anchors and an internal perfect split

Human Compression project: a component of the L019 recovery

An endpoint anchor is a point moved by the first specified involution and fixed by all the others; at the other end the order is reversed. All groups below are actual permutation groups. An eligible edge joins orbits of the actual maximal omission of its colour. We acquire the complete unique-crossing, minimal-transitivity, orbit-budget, saturated forest, and split-inheritance arguments, with their exact hypotheses. In particular, no restriction or block quotient is assumed to inherit the intersection property.

Lemma 1. *Suppose commuting groups L, R have transitive product on A. If R fixes a point $e \in A$, then L is transitive on A and R fixes A pointwise.*

Proof. Every element of the product sends e to a point of e^L, since R fixes e and commutes with L. Thus $e^L = A$. Every element of R fixes every point e^l, again by commutation. □

Lemma 2. *Let a connected transitive string action of rank t on d points have all maximal omissions intransitive. If d is odd, $2t = d + 1$, and both orbits of a unique crossing admit nontrivial block systems for its omission, that crossing is perfect.*

Proof. Write their block numbers and sizes as $m_1, k_1, m_2, k_2 \geq 2$. The complete orbit-budget argument gives

$$t \leq m_1 + m_2 + k_1 + k_2 - 3, \qquad 2t \leq d + 2 - D, \quad D = \sum_{j=1}^{2} (m_j - 2)(k_j - 2).$$

Here $D \geq 0$ and $D \equiv d \pmod 2$, since $m_j k_j$ is the size of the corresponding orbit. Therefore $D \geq 1$. As $2t = d+1$, equality holds throughout and $D = 1$. For an inclusion-minimal subfamily M transitive on the two block sets, let c_j be its number of point orbits on side j. The inequalities proving the displayed bound are precisely

$$|M| \leq m_1 + m_2 - 2, \qquad c_j \leq k_j, \qquad t - 1 - |M| \leq c_1 + c_2 - 2.$$

They must all be equalities.

These are exactly the hypotheses of the constructive part of the acquired saturated-forest proof. To identify its interface explicitly: the block images of M are the two natural transposition paths; the crossing blocks are endpoints, so their labels occupy the two intervals next to the crossing. Every remaining mark fixes the crossing points, centralizes a full block path or its far-end deletion, and therefore has trivial image on each original block set. Each point orbit of $\langle M \rangle$ meets each block once. The selected eligible edges of the remaining marks form two trees on these point orbits. Any additional edge passing across a different selected colour would contradict that colour's ambient eligibility. Thus every remaining mark acts on only one original side. The marks of M already do so. Connectedness of the noncommutation diagram then puts all earlier marks on one side and all later marks on the other. This is a perfect split. The acquired proof after the equality step uses only the three saturated inequalities above, not the former numerical hypothesis $2t = d + 2$; hence it applies unchanged here. □

Theorem 3. *Let $H = \langle s_0, \ldots, s_{t-1} \rangle$ be a transitive string action with connected noncommutation diagram and all maximal omissions intransitive. Suppose it has endpoint anchors at both ends and $2t > d$, where d is its degree. Then it has a perfect split.*

Proof. If $2t > d+1$, the acquired boundary theorem applies. It remains only $2t = d+1$ with d odd. The paired-cut theorem excludes all colours having two eligible edges, so choose a unique crossing at i, with omission orbits A, B.

An endpoint unique crossing with a singleton orbit is perfect. An interior singleton is impossible here: the first anchor e is moved by s_0, so lies on the nonsingleton side B. The commuting groups $L = H_{<i}$ and $R = H_{>i}$ have transitive product on B, and R fixes e. Lemma 1 makes R fix B pointwise. It fixes the singleton as well, contrary to the presence of its nonidentity marks. Thus both sides are nontrivial unless the cut is already perfect.

If both sides admit nontrivial block systems, Lemma 2 applies. Otherwise a primitive side is available. The acquired primitive-interval argument says that its active indices form an interval next to the cut, except for the degree-two action of the two neighbours; the latter has the acquired bound $2t \le d$ and is excluded. Suppose the cut is not perfect. Reverse the order if necessary so that a shared mark lies to the right. The primitive interval then lies to the right, so name its side B; all marks of $L = H_{<i}$ fix B, while $R = H_{>i}$ is transitive there. If $i = 0$, the first anchor is fixed by the whole omission, giving a singleton orbit, a case already settled. If $i > 0$, the first anchor lies in A, because $s_0 \in L$ moves it and fixes B. The commuting product LR is transitive on A and R fixes this anchor. Lemma 1 implies that R fixes A pointwise. There is no shared right mark, a contradiction. The reversed orientation uses the other endpoint anchor. Every possible unique crossing has therefore supplied a perfect split. □

Theorem 4. *Let a string C-marking generate the natural S_n, with $n \ge 7$, rank r, and $2r > n+2$. Then it has a perfect split whose index belongs to $\{1, \dots, r-2\}$.*

Proof. The acquired initial-fracture theorem makes every maximal omission intransitive, and the acquired strict theorem supplies a perfect split. The noncommutation diagram is connected: its components would generate commuting normal factors of S_n, whereas every nontrivial normal subgroup contains A_n and its centralizer is trivial. If the split is internal there is nothing to prove. Reverse if needed to put it at index 0.

Its actual right wing has degree $n-1$ and rank $r-1$. The acquired wing theorem proves, without using its final strict inequality, that every nonsymmetric wing has rank at most half its degree. Here $r-1 > (n-1)/2$, so the wing is the natural S_{n-1}. It has connected noncommutation diagram, inherits the omission hypothesis, and satisfies $2(r-1) > (n-1)+1$. The boundary theorem gives a perfect split at some index $j \ge 1$. If $j = 1$, the adjacent-inheritance lemma applies since $2(r-1) > n-2$ and supplies an internal parent split. If $j \ge 2$, the nonadjacent-inheritance lemma transfers it to the parent, including $j = r-1$. Thus the only case left has perfect splits at both 0 and $r-1$.

Let their singleton endpoints be a, z. They are distinct: a is moved by ρ_0 and fixed by ρ_{r-1}, whereas z is moved by ρ_{r-1} and fixed by ρ_0. The two actual wings have rank $r-1 > (n-1)/2$, so are respectively $\mathrm{Sym}(\Omega \setminus \{a\})$ and $\mathrm{Sym}(\Omega \setminus \{z\})$. The actual intersection property gives

$$T = \langle \rho_1, \dots, \rho_{r-2} \rangle = \mathrm{Sym}(\Omega \setminus \{a, z\}).$$

This proves transitivity of the middle group, rather than assuming it from the two endpoint cuts. It has degree $d = n-2 \ge 5$, rank $t = r-2$, and connected noncommutation diagram. Inside either wing, the other endpoint remains a perfect split because T is transitive on the complement of its singleton. Applying the wing omission lemma twice shows that all maximal omissions of T are intransitive.

Write the outer crossing edges as $\{a, b\}$ and $\{c, z\}$. Commutation of the outer marks shows that each fixes the other's inner endpoint; in particular $b, c \notin \{a, z\}$. Every middle mark other than ρ_1 fixes b, and every one other than ρ_{r-2} fixes c, by the unique-crossing relations. Transitivity of T forces the respective first and last marks to move them. Thus they are endpoint anchors of the actual middle action. Since $2t = 2r-4 > n-2 = d$, Theorem 3 gives a perfect split at an index $k \in \{1, \dots, r-2\}$.

For $2 \leq k \leq r-3$, transfer this split through the two endpoint cuts using nonadjacent inheritance. If $k = 1$, first transfer it through the last endpoint to the wing omitting ρ_0; it is nonadjacent to that endpoint since $r \geq 5$. Then use adjacent inheritance through the first endpoint, again with $2(r-1) > n-2$. The case $k = r-2$ is its reverse. Every case gives an internal perfect split of the parent. □

Attribution and scope. The internal-split conclusion is the corresponding step of Section 8 of Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. The complete earlier rank theorems, actual wing bounds, fracture and saturated-forest proofs remain attributed acquired inputs. The new endpoint-anchor argument replaces the source's exceptional no-perfect-split diagram classification at the odd middle boundary. The use of the actual intersection property to identify the middle symmetric group and the changed saturation hypotheses are explicit. This does not yet prove a trivial internal factor, the leaf-kernel conditions, or the reversible counting correspondence. No whole-original or new-conjecture credit is granted.